



State reconstruction in a nonlinear vehicle suspension system using deep neural networks

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Abstract Modern control synthesis assumes that observability of a dynamic system is satisfied. However, in particular, observability may not be met due to manufacturing cost. To cope with this challenging dilemma, we consider training a neural network with a large dataset that contains prior information of a given nonlinear vehicle dynamics system. In this paper, we proposed and designed a long short-term memory (LSTM)-based neural network to estimate the velocity and position states of a full car's suspension system using only online data streams from cheap inertial sensory measurements. In the training stage, we collect the input and output data that we want to reconstruct using a nonlinear full car model simulation. An LSTM-based neural network is subsequently trained with the collected data, and it reconstructs the velocity and position states only from the acceleration information inputs in the production stage. Finally, to further enhance the performance via Bayesian filtering, the neural network's outputs are projected into the vehicle dynamics using the extended Kalman filter. To demon-

strate the effectiveness of our approach, we compare the performance of the proposed method against a conventional kinematic Kalman filter. The results show that the proposed method recorded 148.8 times smaller value in terms of the mean squared error (MSE) than that of the benchmark. Furthermore, we analyzed the sensitivity of the proposed neural network with or without gyroscope sensors in the full car model. Then, an illustrative control example for implementing the proposed method is presented. Our approach shows that with only noisy accelerometer and gyroscope sensors, we can successfully reconstruct nearly perfect states in a full car's nonlinear dynamic system using a well-trained LSTM-based neural network combined with the extended Kalman filter.

Keywords Nonlinear systems · Neural network · Multiple degree-of-freedom systems · Vehicle suspension system · Extended Kalman filter

1 Introduction

Vibration control of the suspension systems in automobiles is crucial to ride comfort and passenger safety. By designing an optimal controller, a balance between ride comfort and vehicle maneuverability can be achieved. In this regard, the formulation of full car dynamics includes semi-active suspension components (e.g., dampers) combined with state components. Recently available semi-active suspension con-

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trol strategies include skyhook control, H_∞ controls based on the linear parameter varying (LPV) approach, and model predictive control (MPC) [29].

Such control strategies assume that the observability, or at least detectability, condition required for full car dynamics is met, i.e., the measurement of a set of state variables can be made via sensors to guarantee the observability. However, installing sensors that measure position and velocity is costly in practice [1, 9, 40].

In general, the optimal nonlinear filtering based on Bayesian statistics [6] is to construct a probability density function (PDF) of the state variables using the available measurement data and directly approximate the PDF by solving a Fokker–Planck equation (FPE) [4, 35]. However, calculation of a high-dimensional FPE could be challenging [31, 34], especially in multi-degree-of-freedom systems [19]. Therefore, solving the FPE to estimate the state variables in the vehicle dynamics requires a considerable amount of effort [33].

To deal with such issues, several observer design methodologies that predict the full-state variables of a vehicle using only the acceleration have been proposed. A bilinear observer was proposed in [38] based on the acceleration measurements when an unknown disturbance is present. However, an alternative was not suggested for when the disturbance decoupling condition is not met and only considered the quarter-car model. Authors in [10] proposed an H_∞ observer design methodology that utilizes disturbance decoupling and pole placement using linearized full car model that does not satisfy the observability condition. Although the observer design is possible because it is locally detectable, the incoming disturbance can degrade the performance of the observer on an unobservable mode. As a result, the results in [10] showed limitations on position estimates. In [15], the unscented Kalman filter with the estimated process noise covariance is proposed for the vehicular integrated system using the inertial navigation system and the global positioning system (GPS). Though the results are promising, we only consider acceleration measurements, not the position measurements as GPS.

Some practical approaches include the utilization of a Kalman filter based on the kinematics relationship [16, 30]. However, as these approaches are based on integrating the acceleration, they have limitations that cause the estimations to easily diverge. As mentioned in [30], low-frequency noises are excessively amplified, resulting in distorted state estimates. Although

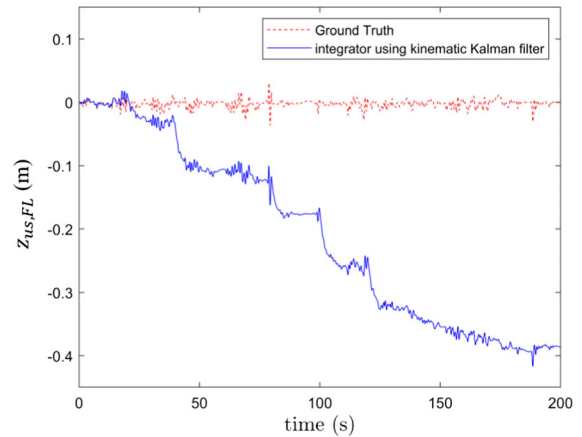


Fig. 1 Integrator using a kinematic Kalman filter

this issue can be mitigated via detrending or mean removal, measurement noise in low-frequency still remains within the estimated signals in practice. In Fig. 1, the results of a full car suspension system example show the estimation of the position value of the tire position ($z_{us, FL}$) using the acceleration information $\ddot{z}_{us, FL}$ via a kinematic Kalman filter (KKF). The KKF, utilized as a part of our benchmark in Sect. 3.5, easily diverges especially when there is an exogenous disturbance as suggested in [30].

Alternatives focus on utilizing artificial neural networks to approximate the nonlinear function mapping through supervised learning. In a Bayesian approach, integration over the posterior of the weight space in neural networks is utilized to predict the outputs based on the new inputs [37]. The posterior distribution could be updated using the prior distribution, which is obtained from the training data [23], i.e., a large dataset that is provided to train the neural networks offline.

Authors in [5] first introduced neural networks for modeling nonlinear time series, and [32] and [7] used a recurrent neural network (RNN) that captures the sequential patterns of a nonlinear state-space system if the system is deterministic. Even so, the use of an RNN involves the training of a large number of parameters and can be inefficient [39]. A recent study about the state-space long short-term memory (LSTM) model showed that it is effective in the sequential modeling of a dynamic system in terms of interpretability [25, 42]. In [27], a radial basis function neural network is employed to estimate the unmodeled uncertainties on the basis of an adaptive extended Kalman filter (EKF).

The contributions of the paper are as follows. The objective of this paper is to estimate the velocity and position states of a full car's nonlinear suspension system using only cheap inertial sensory measurements. To this end, we proposed and designed an LSTM-based neural network which reconstructs nearly perfect states in the (nonlinear) suspension system. Additionally, we successfully trained our neural network model without the observability condition by using a big data as a prior distribution. With model of the vehicular suspension system, we showed how to further enhance the performance via Bayesian filtering by projecting the outputs of the neural network into the vehicle dynamics using the extended Kalman filter. To the best of the author's knowledge, this is the first time that such an LSTM network has been developed and implemented for estimating states only using cheap inertial sensors such as accelerometers with or without gyroscope sensors. As the LSTM-based state estimator based on such inertial measurements is inexpensive with its possible applications to a variety of vehicles, the successful results of the proposed approach have a significant impact to the automotive industry. To further enhance our approach's practicality, we showed how to train the proposed neural network in practice by differentiating the selection of the inertial sensors in the training stage and the production stage (Fig. 2). Next, we demonstrated the effectiveness of our approach by comparing the performance of the state reconstruction of the proposed neural network against a conventional kinematic Kalman filter. Furthermore, we analyzed the sensitivity of the proposed neural network with or without gyroscope sensors in the full car model. Finally, we implemented the proposed method to the linear-quadratic regulator (LQR) controller as an illustrative example of controlling semi-active suspension system.

The paper is organized as follows: In Sect. 2, our problem is formulated after introducing a full car suspension system with model parameters. In Sect. 3, the methods are described. In Sect. 4, the results of the proposed simulations are presented. Finally, the conclusion in Sect. 6 follows the discussion in Sect. 5.

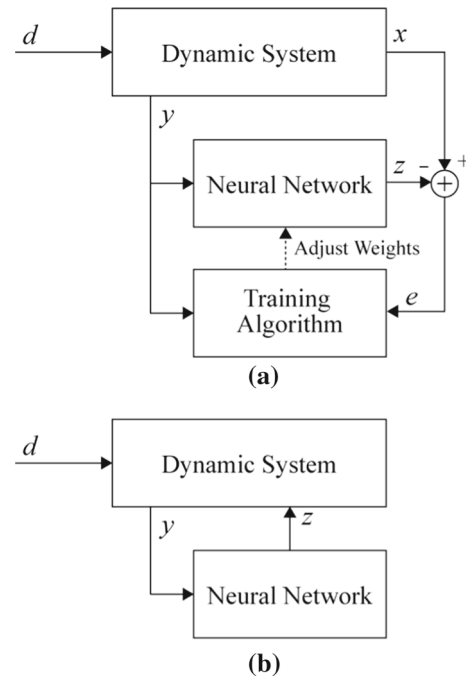


Fig. 2 d is the road input to the control system, y is the acceleration measurement with noise, x is the velocity and displacement states without noise, and z is the neural network output, which is the estimated velocity and displacement states. **a** Describes the data flow, while the neural network is training; a training algorithm in (a) adjusts the weights in the neural network by using a backpropagation algorithm; **b** describes the data flow in the production stage which is after when the neural network is trained

2 Problem definition

2.1 Target system

Our target system is a full car suspension system with semi-active suspensions as the actuators. We utilized the nonlinear full car model of [43] as follows.

$$\dot{x} = f(x, u, z_r) \quad (1)$$

where

$$\begin{aligned}
x &= [z_{COM}, \dot{z}_{COM}, \phi, \dot{\phi}, \theta, \dot{\theta}, \\
&\quad z_{us,FL}, \dot{z}_{us,FL}, z_{us,FR}, \dot{z}_{us,FR}, z_{us,RL}, \\
&\quad \dot{z}_{us,RL}, z_{us,RR}, \dot{z}_{us,RR}]^T \\
&= [x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8, x_9, x_{10}, \\
&\quad x_{11}, x_{12}, x_{13}, x_{14}]^T,
\end{aligned}$$

is the system state, $u = [u_{FL}, u_{FR}, u_{RL}, u_{RR}]^T$ is the control input, and $z_r = [z_{r,FL}, z_{r,FR}, z_{r,RL}, z_{r,RR}]^T$ is the road disturbance. Applying force and torque balance equations to the full car model leads to the following nonlinear state-space equation for the static equilibrium positions.

$$\begin{aligned}
\dot{x}_1 &= x_2, \\
\dot{x}_2 &= \frac{1}{m_s} (-F_{s,FL} - F_{s,d,FL} - F_{s,FR} - F_{s,d,FR} \\
&\quad - F_{s,RL} - F_{s,d,RL} - F_{s,RR} - F_{s,d,RR}), \\
\dot{x}_3 &= x_4, \\
\dot{x}_4 &= \frac{1}{I_{xx}} \left((-F_{s,FL} - F_{s,d,FL} + F_{s,FR} \right. \\
&\quad \left. + F_{s,d,FR}) \frac{s}{2} \cos \phi \right. \\
&\quad \left. + (-F_{s,RL} - F_{s,d,RL} + F_{s,RR} \right. \\
&\quad \left. + F_{s,d,RR}) \frac{s}{2} \cos \phi \right), \\
\dot{x}_5 &= x_6, \\
\dot{x}_6 &= \frac{1}{I_{yy}} \left((F_{s,FL} + F_{s,d,FL} + F_{s,FR} \right. \\
&\quad \left. + F_{s,d,FR}) a \cos \theta \right. \\
&\quad \left. - (F_{s,FL} + F_{s,d,RL} + F_{s,RR} + F_{s,d,RR}) b \cos \theta \right), \\
\dot{x}_7 &= x_8, \\
\dot{x}_8 &= \frac{1}{m_{us,FL}} (F_{s,FL} + F_{s,d,FL} - F_{us,FL} - F_{us,d,FL}), \\
\dot{x}_9 &= x_{10}, \\
\dot{x}_{10} &= \frac{1}{m_{us,FR}} (F_{s,FR} + F_{s,d,FR} - F_{us,FR} - F_{us,d,FR}), \\
\dot{x}_{11} &= x_{12}, \\
\dot{x}_{12} &= \frac{1}{m_{us,RL}} (F_{s,RL} + F_{s,d,RL} - F_{us,RL} - F_{us,d,RL}), \\
\dot{x}_{13} &= x_{14}, \\
\dot{x}_{14} &= \frac{1}{m_{us,RR}} (F_{s,RR} + F_{s,d,RR} - F_{us,RR} - F_{us,d,RR}).
\end{aligned} \tag{2}$$

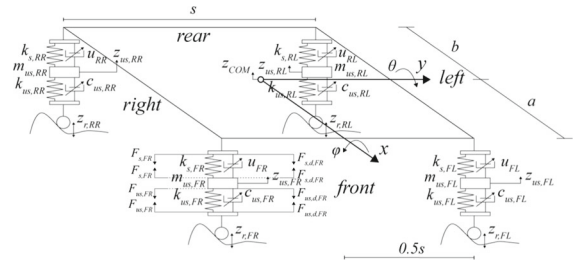


Fig. 3 Full car diagram

where m_s is the sprung mass, m_{us} is the unsprung mass, i.e., tire, I_{xx} is the mass moment of inertia with respect to x axis, and I_{yy} is the mass moment of inertia with respect to y axis.

The forces of the sprung masses in (2) are given by

$$\begin{aligned}
F_{s,p} &= k_{s,p} \cdot \Delta_p, \\
F_{s,d,p} &= c_{s,p} \cdot \dot{\Delta}_p - u_p,
\end{aligned} \tag{3}$$

where $F_{s,p}$ is the spring force of the shock absorber and $F_{s,d,p}$ is the damping force of the semi-active suspension. The subscript p represents the position of the springs and dampers (i.e., $p \in \{FL, FR, RL, RR\}$). $k_{s,p}$ and $c_{s,p}$ are the spring constant and the nominal damping constant of a shock absorber at position p , respectively. See Fig. 3 for the forces of the spring and the damper at the FR position. In addition, u_p is the control input given as the damping force at each semi-active suspension in position p . Finally, we define the following symbols for concise presentation.

$$\begin{aligned}
\Delta_{FL} &= \frac{s}{2} \sin \phi - a \sin \theta + z_{COM} - z_{us,FL}, \\
\Delta_{FR} &= -\frac{s}{2} \sin \phi - a \sin \theta + z_{COM} - z_{us,FR}, \\
\Delta_{RL} &= \frac{s}{2} \sin \phi + b \sin \theta + z_{COM} - z_{us,RL}, \\
\Delta_{RR} &= -\frac{s}{2} \sin \phi + b \sin \theta + z_{COM} - z_{us,RR}, \\
\dot{\Delta}_{FL} &= \frac{s}{2} \dot{\phi} \cos \phi - a \dot{\theta} \cos \theta + \dot{z}_{COM} - \dot{z}_{us,FL}, \\
\dot{\Delta}_{FR} &= -\frac{s}{2} \dot{\phi} \cos \phi - a \dot{\theta} \cos \theta + \dot{z}_{COM} - \dot{z}_{us,FR}, \\
\dot{\Delta}_{RL} &= \frac{s}{2} \dot{\phi} \cos \phi + b \dot{\theta} \cos \theta + \dot{z}_{COM} - \dot{z}_{us,RL}, \\
\dot{\Delta}_{RR} &= -\frac{s}{2} \dot{\phi} \cos \phi + b \dot{\theta} \cos \theta + \dot{z}_{COM} - \dot{z}_{us,RR}.
\end{aligned} \tag{4}$$

The forces of the unsprung masses in (2) are as follows:

$$\begin{aligned} F_{us,p} &= k_{us,p} \cdot \Delta_{us,p}, \\ F_{us,d,p} &= c_{us,p} \cdot \dot{\Delta}_{us,p}, \end{aligned} \quad (5)$$

where $F_{us,p}$ is the spring force of the tire, $F_{us,d,p}$ is the damping force of the tire. $k_{us,p}$ and $c_{us,p}$ are the spring constant and the damping constant of the tire at position p , respectively. In addition,

$$\Delta_{us,p} = z_{us,p} - z_{r,p},$$

where the subscript r indicates the road condition. Note that we assume that the spring constants, the damping constants, and the unsprung masses are all identical regardless of position p . The system parameters of the vehicle model used in this study are listed in Table 1. The spring and damping constants are assumed to be identical for all positions p .

Note that there is a Kalman-filter-based observer design with the output given by an acceleration [3, 13, 36] in a quarter car system. In a quarter car, the observability condition is met, i.e., the rank of the observability matrix is the same as the number of states. Similar approach is also possible for linearized full car system using the observer design methodology in [10]. However, the observability condition is not met in full car system, i.e., the rank of the observability matrix using the approach proposed in [10] is 12.

Table 1 Vehicle model parameters

Parameter	Description	Value
m_s	Sprung mass	1000 kg
$m_{us,p}$	Unsprung mass (tires)	50 kg
I_{xx}	Mass moment of inertia on x -axis	150 kgm ²
I_{yy}	Mass moment of inertia on y -axis	346.875 kgm ²
a	Length from front of body to COM*	0.975 m
b	Length from rear of body to COM*	0.975 m
s	Width of body	1.2 m
$k_{s,p}$	Spring constant of shock absorbers	2.1 kgf/mm
$c_{s,p}$	Nominal damping constant of shock absorbers	10 Ns/m
$k_{us,p}$	Spring constant of tires	30 kgf/mm
$c_{us,p}$	Damping constant of tires	50 Ns/m

COM*: Center of mass

The ISO-8608 standard is adopted to produce realistic road conditions in the simulation, as in [20]. Throughout the study, the generated road is composed of class-B and class-C road in accordance with ISO-8608 at a 1:1 ratio. In addition, randomly generated speed bumps are included in the road set. Different road sets are given to the left and right tires to create more adverse conditions. The properties of the semi-active actuators can be found in [28] and [29].

2.2 Problem statement

Measuring the velocity and position values of the suspension system is costly. For economical vehicle manufacturing, we only focus on utilizing cheap inertial sensors. Therefore, we assume that the observation values are given only in the forms of the acceleration and angular velocity, which implies that the states are not observable. Under the assumptions, our problem is to design a novel estimation system that can estimate the full-state values x in (1). In consideration of the passenger comfort, the estimated full-state values are fed into a controller to minimize the vertical acceleration applied to the sprung mass. Accelerometers and a six-axis gyroscope are used as sensors. The accelerometers are assumed to be attached to the centers of the four tires and four polytopes of the sprung mass (immediately above the four tires), while the gyroscope is attached to the center of the sprung mass. The accelerometers provide heave acceleration values, and the gyroscope provides the heave acceleration and angular velocities of the body mass.

A linear quadratic regulator (LQR) is utilized as a controller. Since the LQR controller cannot deal directly with the acceleration, we minimize the velocity and position values of the roll, pitch, and heave motion, i.e., z , $\dot{z}$, ϕ , $\dot{\phi}$, θ , and $\dot{\theta}$, instead. Note that owing to the nonlinearities of full car dynamics and semi-active actuators, the LQR can be regarded as a suboptimal controller.

3 Methods

In this section, we explain the method used for collecting data, and preprocessing procedures for the data collected from the full car model. Subsequently, we describe the structure of the proposed neural network

for reconstructing the full-state values, design of the EKF, and benchmark model.

3.1 Data collection

We set up a simulation in MATLAB utilizing the full car model (Fig. 3) and generated road disturbances described in Sect. 2. The simulation records the dynamics of the vehicle model via the LQR controller under the road disturbance during each 0.01 s of simulation time as described in Fig. 2a. The angular velocity and displacement data are collected at the center of mass. Additionally, the z -direction acceleration, velocity, and displacement information are collected at the corners of the vehicle and at the center of mass. We collected 2,000,000 timesteps of data using 1,000 randomly created ISO-8608-based roads. The car model in the simulation drives each road for 20 s.

As the required state space is composed of the velocity and position information, we classify the collected data as the input data and the ground truths as the output results of the neural network. The input data consists of the vehicle longitudinal velocity and the information measured from the accelerometers and gyroscope, which is

$$\text{inputs} = [\ddot{z}_{s,FL}, \ddot{z}_{us,FL}, \ddot{z}_{s,FR}, \ddot{z}_{us,FR}, \ddot{z}_{s,RL}, \ddot{z}_{us,RL}, \ddot{z}_{s,RR}, \ddot{z}_{us,RR}, \ddot{z}_{COM}, \dot{\phi}, \dot{\theta}, v_{car}].$$

Here, $\ddot{z}_{s,FL}$, $\ddot{z}_{us,FL}$, $\ddot{z}_{s,FR}$, $\ddot{z}_{us,FR}$, $\ddot{z}_{s,RL}$, $\ddot{z}_{us,RL}$, $\ddot{z}_{s,RR}$ and $\ddot{z}_{us,RR}$ are measured from the accelerometers; $\ddot{z}_{COM}$, $\dot{\phi}$ and $\dot{\theta}$ are measured from the six-axis gyroscope.

The output results from the neural network are

$$\text{outputs} = [\dot{z}_{s,FL}, z_{s,FL}, \dot{z}_{us,FL}, z_{us,FL}, \dot{z}_{s,FR}, z_{s,FR}, \dot{z}_{us,FR}, z_{us,FR}, \dot{z}_{s,RL}, z_{s,RL}, \dot{z}_{us,RL}, z_{us,RL}, \dot{z}_{s,RR}, z_{s,RR}, \dot{z}_{us,RR}, z_{us,RR}, \dot{z}_{COM}, z_{COM}, \phi, \theta].$$

Subsequently, the paired data are split into three groups: 70% for training, 20% for evaluation, and 10% for testing.

The acceleration and angular velocity values of the inputs are contaminated with the noise of the standard

deviation of the *MPU-6050* sensor, which is considered the epitome of cheap accelerometers and gyroscopes [24].

3.2 Preprocessing

Each data column of the collected dataset has a different range, which may inhibit the neural network's ability to fit its weight variables to estimate the proper control states. Therefore, we normalize the data by removing the mean and scaling it to the unit variance as follows.

$$z = \frac{x - \mu}{s},$$

where μ is the mean of each column in the training dataset, and s is the standard deviation of each column in the training dataset.

In addition, the dynamics of the system change with respect to time as a disturbance signal feeds into the system. We stacked the data from the window-length (w in Fig. 4) timesteps before to the current timestep. By stacking the data in a time series, the neural network is able to extract the time sequential features.

3.3 Network structure

The input data to the neural network are divided into two groups: the sensor data and the forward velocity information of the vehicle. The sensor data, stacked for window-length w timesteps, consist of the acceleration information and angular velocity information and are described as follows:

$$x_{\text{sensor}} = \begin{bmatrix} \ddot{z}_{s,t_n} & \ddot{z}_{us,t_n} & \ddot{z}_{COM,t_n} & \dot{\phi}_{t_n} & \dot{\theta}_{t_n} \\ \ddot{z}_{s,t_{n-1}} & \ddot{z}_{us,t_{n-1}} & \ddot{z}_{COM,t_{n-1}} & \dot{\phi}_{t_{n-1}} & \dot{\theta}_{t_{n-1}} \\ \ddot{z}_{s,t_{n-2}} & \ddot{z}_{us,t_{n-2}} & \ddot{z}_{COM,t_{n-2}} & \dot{\phi}_{t_{n-2}} & \dot{\theta}_{t_{n-2}} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \ddot{z}_{s,t_{n-w+1}} & \ddot{z}_{us,t_{n-w+1}} & \ddot{z}_{COM,t_{n-w+1}} & \dot{\phi}_{t_{n-w+1}} & \dot{\theta}_{t_{n-w+1}} \end{bmatrix},$$

where $\ddot{z}_{s,t_n}$, $\ddot{z}_{us,t_n}$, and $\ddot{z}_{COM,t_n}$ are the acceleration information in the z -direction at timestep t_n on the sprung mass, unsprung mass, and center of mass, respectively. $\dot{\phi}$ and $\dot{\theta}$ are the angular velocity information along the x - and y -axes, respectively.

The forward velocity information of a vehicle stacked for window-length w timesteps is described

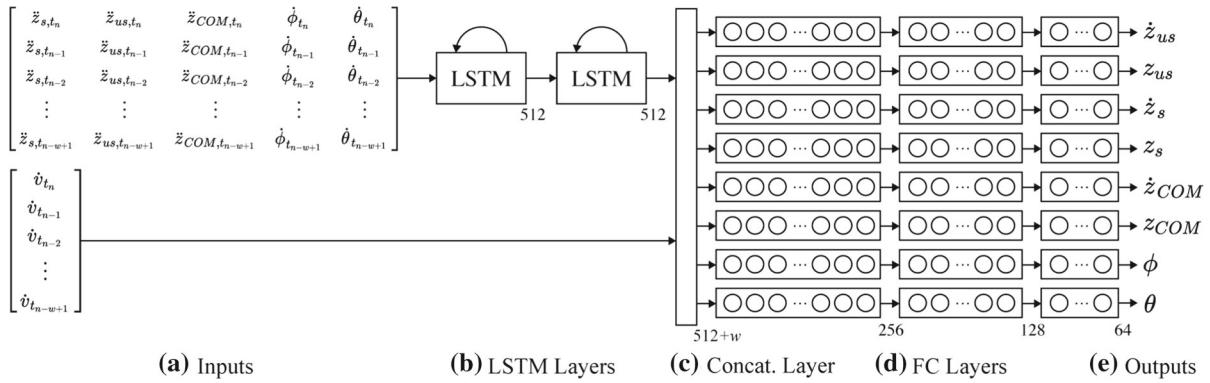


Fig. 4 The input data consist of measurements made by the accelerometers and gyroscope are stacked with a window length of w . The stacked velocity information of a vehicle skips the long short-term memory (LSTM) layers and is fed directly into the concatenated layer. **b**) The LSTM layer is capable of extracting temporal features from the sequential data. The upward arrows

indicate the information that sequentially travels through the cells in the LSTM layer. **c** The features extracted from the LSTM layers and the vehicle velocity information are concatenated and fed into the fully connected shared layers. **d, e** The network is split into 8 branches, which estimate $\dot{z}_s$, z_s , $\dot{z}_{us}$, z_{us} , $\dot{z}_{COM}$, z_{COM} , ϕ , and θ

as follows:

$$x_{v_{car}} = \begin{bmatrix} v_{car,t_n} \\ v_{car,t_{n-1}} \\ v_{car,t_{n-2}} \\ \vdots \\ v_{car,t_{n-w+1}} \end{bmatrix},$$

where v_{car,t_n} is the forward direction velocity information of a vehicle at timestep t_n .

The stacked data are fed into the LSTM [14] layers. The LSTM, which solved a vanishing gradient problem [17], is an improvement over the general RNN. The LSTM network is capable of extracting temporal features from the sequential data. The temporal features are stored and carried forward by using cells. In contrast to the RNN, the LSTM adds input-, forget-, and output-gates to the hidden layer such that it can delete unnecessary memories and select the memories to retain among the cells. The gates and cell states in the LSTM are described in Fig. 5 and defined as follows:

$$\begin{aligned} i_t &= \sigma(W_{x_i}x_t + W_{h_i}h_{t-1} + b_i) \\ g_t &= \tanh(W_{x_g}x_t + W_{h_g}h_{t-1} + b_g) \\ f_t &= \sigma(W_{x_f}x_t + W_{h_f}h_{t-1} + b_f) \\ C_t &= f_t \odot C_{t-1} + i_t \odot g_t \\ o_t &= \sigma(W_{x_o}x_t + W_{h_o}h_{t-1} + b_o) \\ h_t &= o_t \odot \tanh(c_t), \end{aligned}$$

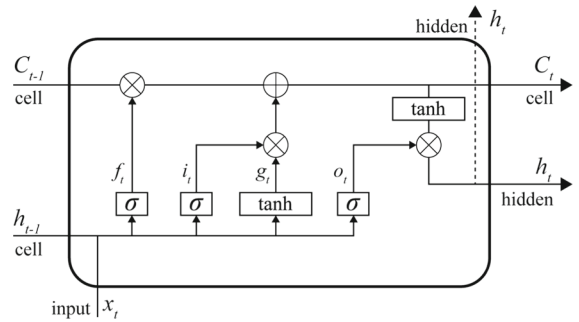


Fig. 5 LSTM block architecture

where σ is a logistic sigmoid function. W_{x_i} , W_{x_g} , W_{x_f} , and W_{x_o} are the weight vectors used in each gate with x_t . W_{h_i} , W_{h_g} , W_{h_f} , and W_{h_o} are the weight vectors used in each gate with h_{t-1} . b_i , b_g , b_f , and b_o are the bias vectors in each gate. $\odot$ is the Hadamard product, which is the element-wise product operation.

Subsequently, the features extracted from the sensor data and the velocity information of the vehicle are concatenated as described in Fig. 4c. Afterward, the network is split into 8 branches that share the features from the previous layers. The 8 branches estimate $\dot{z}_s$, z_s , $\dot{z}_{us}$, z_{us} , $\dot{z}_{COM}$, z_{COM} , ϕ , and θ , respectively. The splitting of the neural network branches ensures less inference between the decisions for each state estimation that are made by the fully connected layers.

Each branch produces different outputs with a different size: 8 outputs for $\dot{z}_s$, 8 outputs for z_s , 8 outputs for $\dot{z}_{us}$, 8 outputs for z_s , 1 output for $\dot{z}_{COM}$, 1 output for z_{COM} , 1 output for ϕ , and 1 output for θ .

The neural network is trained using the Adam optimizer [18] as a part of the training algorithm in Fig. 2a. The Adam optimizer is a first-order gradient-based optimizer which is computationally efficient, while it works well compared to other adaptive learning-method algorithms. The mean squared error (MSE) is selected as a loss function, and the training algorithm reduces the learning rate by multiplying 0.05 to its previous learning rate when there is no decrease in the MSE values for 10 epochs. Finally, the training algorithm stops the training when there is no decrease in the MSE values for 20 epochs even after reducing the learning rates.

3.4 Extended Kalman filter

We employed the EKF to project the output of the neural network onto the full car dynamics. The full car dynamics model is discretized using the forward difference method with a sampling period of $T = 0.01$ s to design the EKF. The overall process of the EKF is summarized as follows:

- Initialize the predicted error state estimate $\hat{x}_{k|k-1}$ and predicted error covariance $P_{k|k-1}$.
- Set the process noise covariance matrix Q and observation noise covariance matrix R .
- Perform the following actions during the iterations:
 1. Obtain the full-state observations z_k from the neural network.
 2. Compute the Kalman gain using $K = P_{k|k-1}C^T(CP_{k|k-1}C^T + R)^{-1}$
 3. Update the state estimate using $\hat{x}_{k|k} = \hat{x}_{k|k-1} + K(z_k - C\hat{x}_{k|k-1})$
 4. Update the estimated covariance using $P_{k|k} = (I - KC)P_{k|k-1}$
 5. Update the predicted state estimation and predicted error covariance using $\hat{x}_{k+1|k} = g(\hat{x}_{k|k}, u_k)$, $P_{k+1|k} = F_k P_{k|k} F_k^T + Q$

where F_k is $F_k = \frac{\partial f}{\partial x}$.

- The filtered observation is given by $\hat{x}_{k|k}$.

Table 2 Noise levels for EKF

Parameter	Value	Parameter	Value
q_1	6.8027	r_1	8.8680×10^{-6}
q_2	9.5347	r_1	1.9472×10^{-4}
q_3	0.2072	r_1	2.4786×10^{-5}
q_4	0.1	r_4	1.0128×10^{-4}
q_5	7.0183	r_5	4.1939×10^{-7}
q_6	0.1	r_6	1.0056×10^{-4}
q_7	0.1694	r_7	7.4184×10^{-7}
q_8	0.1	r_8	8.0474×10^{-4}
q_9	7.6593	r_9	1.6603×10^{-6}
q_{10}	0.1	r_{10}	7.6539×10^{-4}
q_{11}	2.3214	r_{11}	3.5284×10^{-5}
q_{12}	0.1	r_{12}	9.9906×10^{-4}
q_{13}	1.8936	r_{13}	3.6467×10^{-5}
q_{14}	0.1	r_{14}	1.9560×10^{-3}

The noise covariance matrices Q and R are assumed to be diagonal matrix forms, i.e.,

$$Q = \text{diag}(q_i), \quad R = \text{diag}(r_i), \quad \text{for } i \in \{1, \dots, 14\}. \quad (6)$$

The values of q_i and r_i can be found by solving an optimization problem using the precollected data as follows:

$$q_i = \arg \min_{q_i} \|\hat{x}_{k|k} - x_{gt}\|^2. \quad (7)$$

The values of r_i are determined using

$$r_i = \mathbb{E}[(z_i - x_{i,gt}) - \mathbb{E}[(z_i - x_{i,gt})]]^2. \quad (8)$$

The selection of q_i and r_i is accomplished using one of the datasets from the aforementioned MATLAB simulation. The identified values of q_i and r_i are listed in Table 2.

3.5 Benchmark method

For the benchmark, we consider the KKF assuming the Weiner process acceleration [2, 11, 12]. One major problem here is that there is a diverging term when the acceleration is integrated. Practically, this problem can

be solved by adding a band-pass filter to remove the low-frequency disturbance and high-frequency noise. If the integrated states diverge, then an additional high-pass filter can be adopted. We divided the states x considering the symmetry of the full car dynamics: heave motion at the COM, rolling motion at the COM, pitching motion at the COM, and heave motions of the tires. Then, we designed specific KKF's and band-pass filters for the states. High-pass filters are added when the position is integrated twice from the acceleration.

For the accelerometer-based states, the kinematic relationship is given as

$$\begin{aligned} X_a(k+1) &= \Phi_a X_a(k) + \Gamma_a w(k), \\ Y_a(k) &= H_a X_a(k) + v(k), \end{aligned} \quad (9)$$

where $X_a(k) = [z, \dot{z}, \ddot{z}]^T$, $w \sim \Sigma(0, W)$ is a process noise, $v \sim \Sigma(0, V)$ is a sensor noise,

$$\begin{aligned} \Phi_a &= \begin{bmatrix} 1 & T & 0.5T^2 \\ 0 & 1 & T \\ 0 & 0 & 1 \end{bmatrix}, \quad \Gamma_a = \begin{bmatrix} 0.5T^2 \\ T \\ 1 \end{bmatrix}, \\ H_a &= [0 \ 0 \ 1]. \end{aligned}$$

The process noise covariance matrix Q_a is given in Chapter 6 in [2] as

$$Q_a = \begin{bmatrix} \frac{1}{4}T^4 & \frac{1}{2}T^3 & \frac{1}{2}T^2 \\ \frac{1}{2}T^3 & T^2 & T \\ \frac{1}{2}T^2 & T & 1 \end{bmatrix} W,$$

where W is the process noise variance. On the other hand, the kinematic relationship for the gyroscope-based states is

$$\begin{aligned} X_g(k+1) &= \Phi_g X_g(k) + \Gamma_g w(k), \\ Y_g(k) &= H_g X_g(k) + v(k), \end{aligned} \quad (10)$$

where $X_g(k) = [\vartheta, \dot{\vartheta}]^T$, and

$$\Phi_g = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix}, \quad \Gamma_g = \begin{bmatrix} T \\ 1 \end{bmatrix}, \quad H_g = [0 \ 1].$$

Table 3 Parameters of designed filters for categories of states

Category	$\omega_{b,1}$	$\omega_{b,2}$	ω_h	W	V
COM heave	0.2	25	0.1	100	0.01
Tire heave	1	35	1	100	0.01
Roll	0.1	25	—	0.1585	15.85
Pitch	0.3	25	—	0.6871	1.800

Table 4 Set of sensor data for sensitivity analysis

Model	Input sensor data
Proposed	$\ddot{z}_{s,FL}, \ddot{z}_{s,FR}, \ddot{z}_{s,RL}, \ddot{z}_{s,RR}, \ddot{z}_{us,FL}, \ddot{z}_{us,FR}, \ddot{z}_{us,RL}, \ddot{z}_{us,RR}, \ddot{z}_{COM}, \dot{\phi}, \dot{\theta}$
A	$\ddot{z}_{s,FL}, \ddot{z}_{s,FR}, \ddot{z}_{s,RL}, \ddot{z}_{s,RR}, \ddot{z}_{us,FL}, \ddot{z}_{us,FR}, \ddot{z}_{us,RL}, \ddot{z}_{us,RR}$
B	$\ddot{z}_{COM}, \dot{\phi}, \dot{\theta}$
C	$\ddot{z}_{COM}$
D	$\ddot{z}_{us,FL}$

The process noise covariance matrix Q_g is given as

$$Q_g = \begin{bmatrix} T^2 & T \\ T & 1 \end{bmatrix} W.$$

Again, the noise variances W and v are tuning parameters of the Kalman filter algorithm. We find these tuning parameters by solving an optimization problem based on the Matlab simulation data for each category, in the similar manner as in (7). The design parameters of the band-pass filters are low cutoff frequency ($\omega_{b,1}$) and high cutoff frequency ($\omega_{b,2}$). The design parameters of the high-pass filters are their cutoff frequency (ω_h), and the design parameters of the Kalman filters are the process noise variance (W) and sensor noise variance (V). The parameters of the designed band-pass filters, high-pass filters, and Kalman filters for each category are summarized in Table 3.

4 Simulation design and results

In this section, we describe how we designed and executed the simulations to demonstrate the performance of the proposed method. First, we disabled some of the sensors and trained the neural network to analyze the sensitivity of our proposed neural network model. We compared the estimation error results with neural networks that utilized various sets of sensors.

Then, we gathered the datasets of the states, including the acceleration information from the simulations. Next, we attempted to determine the position and velocity states from the accelerometer and gyroscope data using the proposed method and the benchmark method, i.e., in an offline simulation.

Finally, to demonstrate the implementation, we suggested the LQR control example using reconstructed states from the proposed method and ground truth, respectively, in an online simulation. The road conditions were simulated as suggested in Sect. 2 using ISO-8608. All simulations were performed 50 times to demonstrate the feasibility of the proposed method on average.

4.1 Sensitivity analysis

The neural network designed for this study utilizes a six-axis gyroscope attached to the center of mass and 8 accelerometers located at the corners of the sprung mass and the unsprung mass beneath it. To identify which sensors are important for estimating the velocity and position information from the given sensor inputs, we reduced the number of input sets handled by the input layer of the neural network by disabling some of the sensors in the simulation model.

We divided the inputs into $\ddot{z}_{s,FL}$, $\ddot{z}_{s,FR}$, $\ddot{z}_{s,RL}$, $\ddot{z}_{s,RR}$, $\ddot{z}_{us,FL}$, $\ddot{z}_{us,FR}$, $\ddot{z}_{us,RL}$, $\ddot{z}_{us,RR}$, $\ddot{z}_{COM}$, $\dot{\phi}$, and $\dot{\theta}$. We created four models that utilized different sets of the input data as described in Table 4. We have compared results from the models for different sets of sensors for the same physical state variable vector x . The results are shown in Fig. 6.

4.2 Offline simulation

The performances of the neural network estimator and benchmark estimator are also evaluated using the MSE values. The errors are defined as the difference between the estimated states and ground truth states, and the states are estimated using the proposed method followed by EKF. The results are shown in Table 5 and in Figs. 7, 8, and 9.

Table 5 MSE value comparison

States	MSE of KKF	MSE of Proposed method
x_1	6.9976×10^{-4}	4.7157×10^{-8}
x_2	3.0207×10^{-4}	8.5789×10^{-7}
x_3	1.0619×10^{-4}	8.3940×10^{-8}
x_4	1.0888×10^{-9}	1.1547×10^{-9}
x_5	2.7444×10^{-5}	8.7339×10^{-9}
x_6	3.8333×10^{-9}	4.2820×10^{-9}
x_7	6.8616×10^{-5}	9.7972×10^{-8}
x_8	1.1290×10^{-2}	1.7197×10^{-4}
x_9	3.1560×10^{-5}	6.3739×10^{-8}
x_{10}	9.1305×10^{-3}	1.4147×10^{-4}
x_{11}	5.5905×10^{-5}	1.0039×10^{-7}
x_{12}	5.9760×10^{-2}	9.3217×10^{-5}
x_{13}	3.3555×10^{-5}	7.3338×10^{-8}
x_{14}	5.0428×10^{-2}	1.1725×10^{-4}
Total	5.5820×10^{-3}	3.7517×10^{-5}

4.3 Online simulation

To demonstrate the implementation of the proposed method, a control problem of a semi-active suspension system using the LQR is selected as an example. The LQR is a full-state feedback controller, so it requires a proper observer to estimate all of the states. This cannot be fulfilled only by the accelerometer data. However, since the proposed method can reconstruct the state values from the accelerometer data as described in the production stage in Fig. 2b, full-state feedback can be achieved.

To attenuate the noise from the neural network and to take the dynamics into account, we utilize the EKF as described in Sect. 3. The LQR controller is designed using the following cost function:

$$J_{\text{LQR}} = \frac{1}{2} \int_0^\infty [x^T Q_r x + u^T R u] dt, \quad (11)$$

where

$$Q_r = 400,000 \begin{bmatrix} I_{6 \times 6} & 0_{6 \times 8} \\ 0_{8 \times 6} & 0_{8 \times 8} \end{bmatrix}, \quad R = 0.01 I_{4 \times 4}. \quad (12)$$

Matrix Q_r is selected to focus on controlling the bounce (z_s), roll (ϕ), and pitch (θ) of the sprung mass. To evaluate the performance of the LQR controller, we used

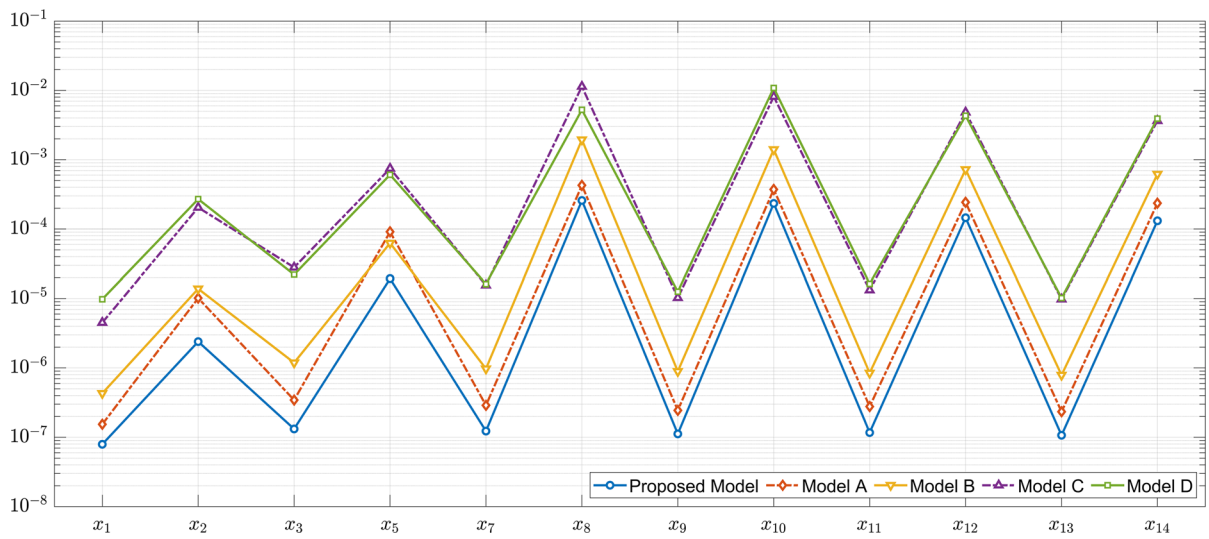


Fig. 6 MSE results for each model in log scale

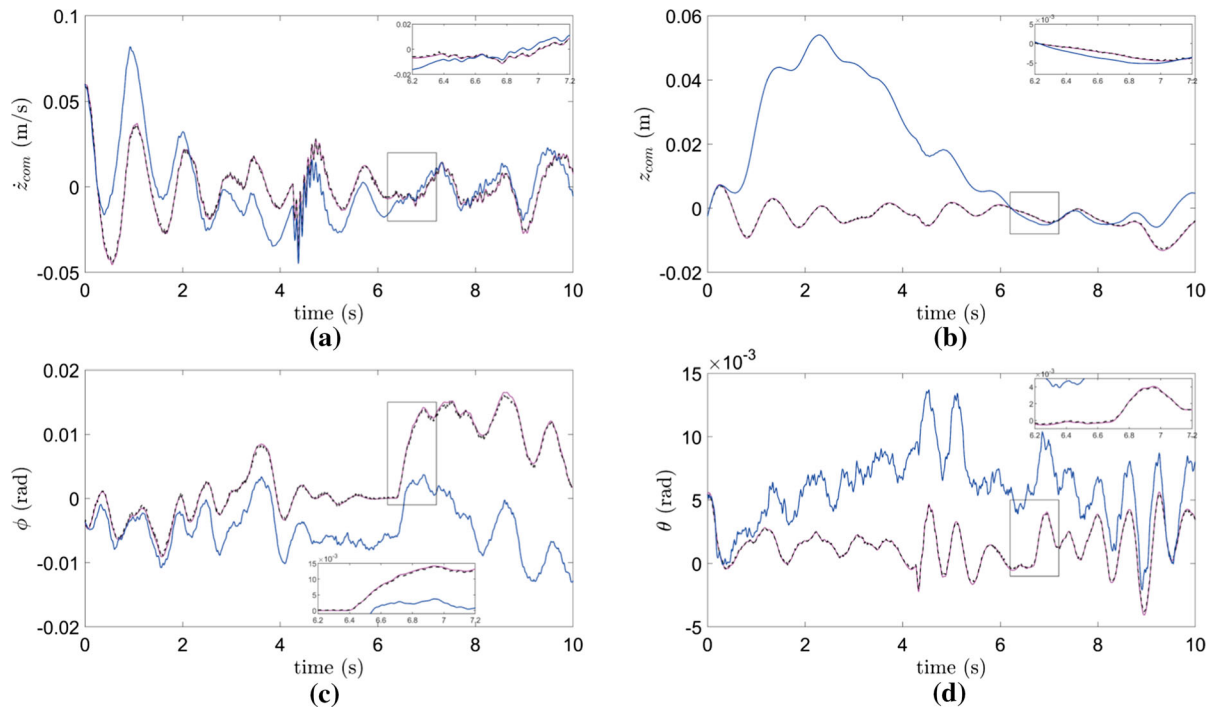


Fig. 7 Estimation results for **a** $\dot{z}_{COM}$, **b** z_{COM} , **c** ϕ , and **d** θ . The estimation results of the proposed model are shown in magenta, the Kalman filter is shown in blue, and the ground truth is shown

by the black dotted line. The ground truth and estimation results of the proposed model lie on top of each other and are therefore difficult to discern

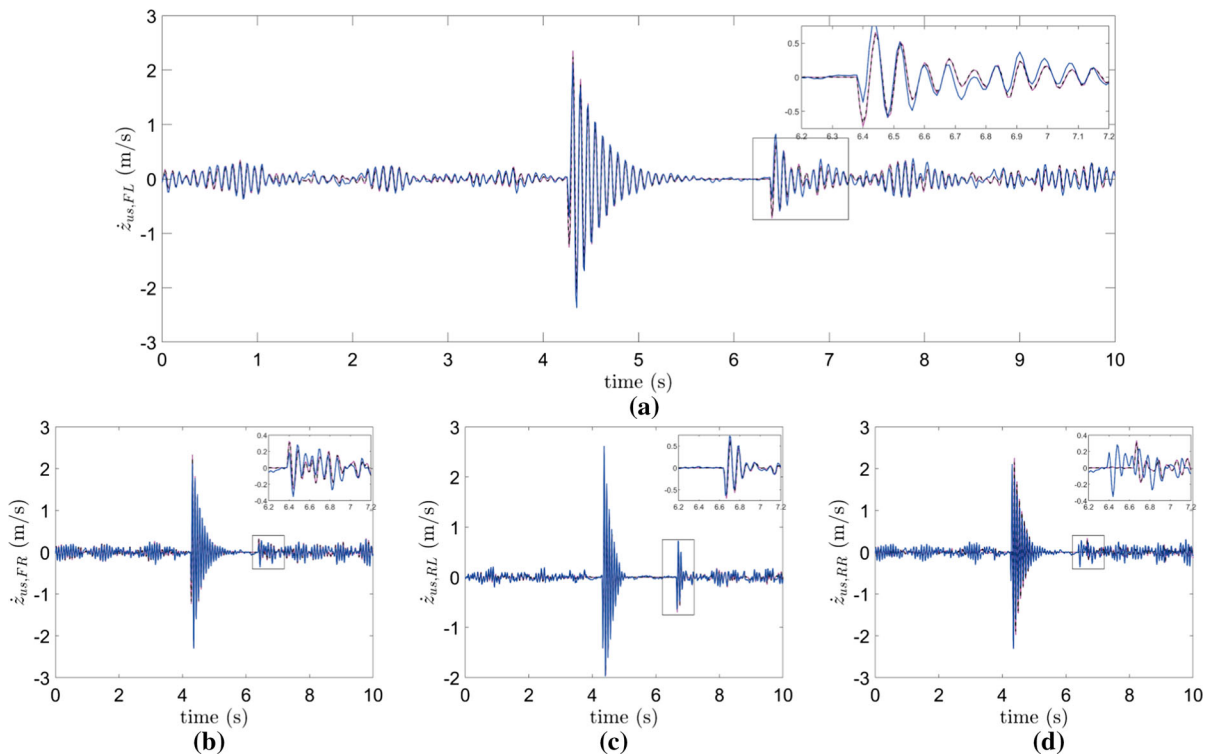


Fig. 8 Estimation results for **a** $\dot{z}_{us,FL}$, **b** $\dot{z}_{us,FR}$, **c** $\dot{z}_{us,RL}$, and **d** $\dot{z}_{us,RR}$. The estimation results of the proposed model are shown in magenta, the Kalman filter is shown in blue, and the ground

truth is shown by the black dotted line. The ground truth and estimation results of the proposed model lie on top of each other and are therefore difficult to discern

the following performance parameter P defined by the average of the root mean square (RMS) value of each state x_i for $i \in \{1 \dots 6\}$, i.e.,

$$P = \frac{1}{6} \sum_{i=1}^6 \text{rms}(x_i). \quad (13)$$

The simulations were performed on 50 different sets of road profiles, but the same 50 sets were used in both cases. The resulting performance parameters are listed in Table 6.

5 Discussions

5.1 Sensitivity analysis

The proposed model utilizes both the z -direction acceleration information and angular velocity information of the vehicle. It also takes the information from both cor-

ners of the vehicle and the information from the center of mass. We selected the set of input sensor data to see whether the neural network was still able to reconstruct the same physical state variable vector x by eliminating sensors from the full list of our proposed sensor placement.

Model A utilizes only the z -direction acceleration information from the corners of the vehicle ($\ddot{z}_s, \ddot{z}_{us}$). Model B utilizes the angular velocity information and z -direction acceleration information at the center of mass ($\ddot{z}_{com}, \dot{\theta}$, and $\dot{\phi}$). In Fig. 6, both model A and model B showed a larger MSE value than the proposed model in estimating the full states x : a 1.74-times larger overall MSE value for model A and 6.03-times larger MSE value for model B. The result indicates that the overall MSE value for model A is smaller than the MSE value for model B, which may be owing to the fact that model A utilizes twice the amount of information than model B does.

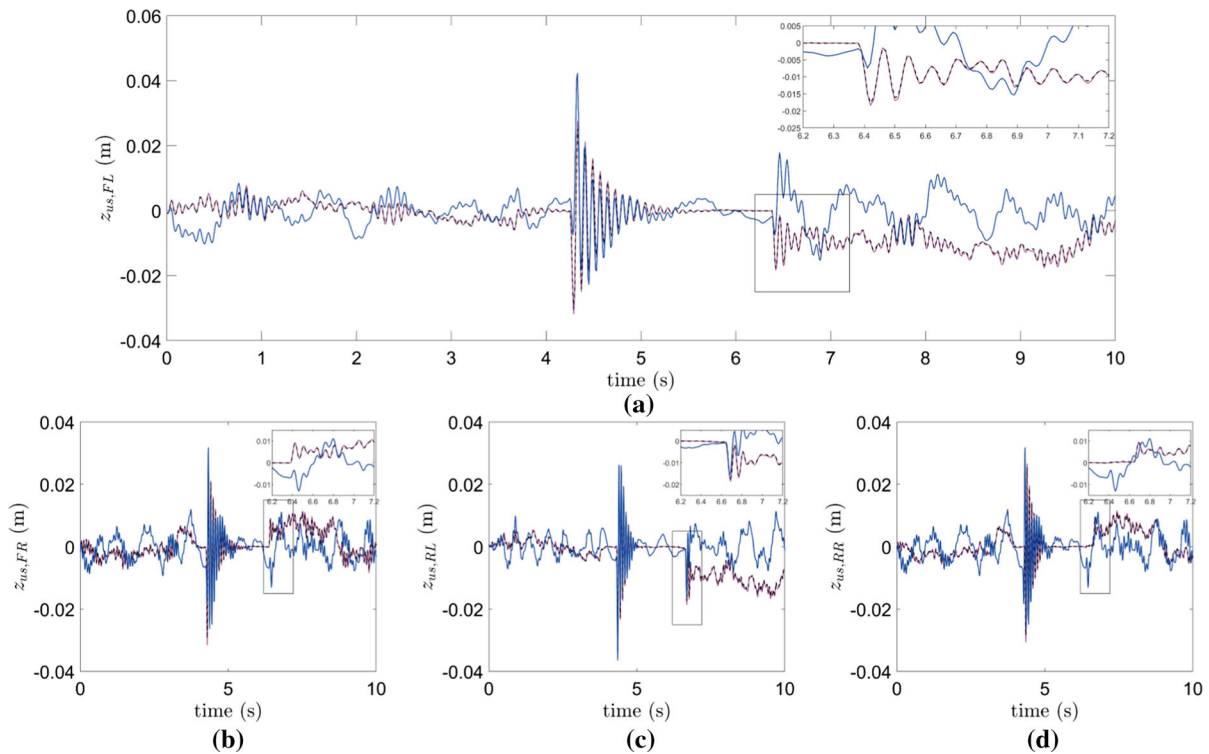


Fig. 9 Estimation results for **a** $z_{us,FL}$, **b** $z_{us,FR}$, **c** $z_{us,RL}$, and **d** $z_{us,RR}$. The estimation results of the proposed model are shown in magenta, the Kalman filter is shown in blue, and the ground

truth is shown by the black dotted line. The ground truth and estimation results of the proposed model lie on top of each other and are therefore difficult to discern

We can assume that the neural network reconstructs the control states only from the given sensory data. Model A performed better than model B when estimating x_5 , which is the pitch of the vehicle model. The neural network successfully reconstructed the information that was needed to estimate the angular displacement of the vehicle because the information at the corners of the vehicle is highly related to the angular states at the center of mass. Please refer to the equation in (4).

Model C and model D utilize only information from a single accelerometer each: the center of mass for model C and the front left body for model D. As a single source of z -direction acceleration information is insufficient to recover the full states, model C and model D recorded poor MSE results compared with the proposed model: an MSE 36.64 times larger for model C and an MSE 31.93 times larger for model D.

Model C performed better than model D when estimating x_1 and x_2 , which are the z -direction displacement and velocity at the center of mass, respectively.

Conversely, model D performed better than model C when estimating x_8 , which is the z -direction displacement at the front left tire ($z_{us,FL}$). These differences may have occurred because even though the single source of information was insufficient to estimate the full states x , the neural network performed better when estimating the state variables at sensor locations with local sensory measurements.

Model C and the model D are the worst scenarios for the proposed neural network structure. A comparison of the results in Fig. 6 reveals that model C and model D performed similarly when estimating each state except for those discussed in the previous paragraph. Therefore, we can infer that the neural network constructed knowledge of the rigid body model by training a large amount of data.

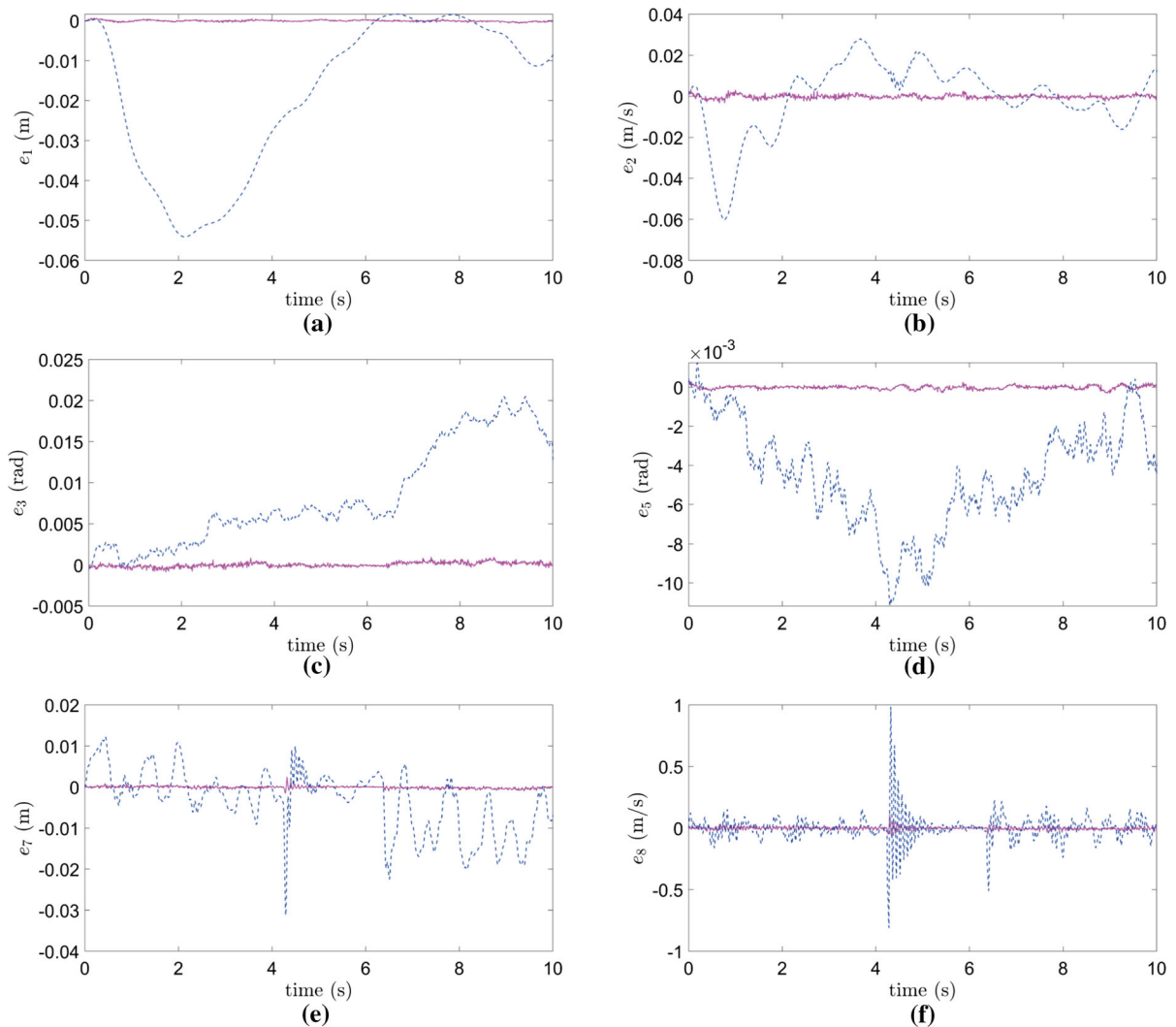


Fig. 10 Estimation errors for **a** z_{COM} , **b** $\dot{z}_{COM}$, **c** ϕ , **d** θ , **e** $z_{us,FL}$, and **f** $\dot{z}_{us,FL}$. The estimation errors of the proposed model are shown in magenta, and the Kalman filter is shown by the blue dotted line

5.2 Offline simulations

In Table 5, it can be seen that the MSE values of the proposed method are smaller by an order of 10^{-2} on average. In Figs. 7, 8, and 9, the estimation of the proposed method reconstructed the states with such accuracy that it was difficult to differentiate from the ground truth.

Note that for the position states at tire positions, i.e., x_7 , x_9 , x_{11} , and x_{13} in Table 5, the estimation of the KKF diverges while the proposed method accurately estimates the states (Fig. 9). This may due to the fact that the disturbances by road conditions are fed to tire

Table 6 Performance parameters for LQR example

State estimation methods	Performance parameters (P)
Ground truth	0.1905
Proposed method + EKF	0.1960

positions directly. Both the proposed and KKF methods estimated the velocity states at tires (x_8 , x_{10} , x_{12} , and x_{14}) well—see Fig. 8. The estimation of the velocity states did not degrade as compared to the position estimate, because the velocity states are integrated once,

and thereby the errors are less likely to accumulate. Similarly, from both methods, the estimation results of x_2 , which are the z -directional velocity states at the center of mass, recorded less MSE values compared to those of x_1 , which is the z -directional position state at the center of mass (Fig. 7).

The estimation error results in Fig. 10 show that the errors of the proposed model are nearly zero as compared to those of the Kalman filter-based approach. The estimation errors of the states which are highly correlated with the impact to the system (θ , $z_{us,FL}$, and $\dot{z}_{us,FL}$) tend to increase when the speed bump hits the full car dynamics, which is around 4.1 seconds in Fig. 10d, e, f. However, the estimation errors of the proposed method stayed near zero as compared to those of the Kalman filter, even after the impact occurred. This shows that the errors do not accumulate when the neural network is used.

The MSE values for the estimations of x_4 and x_6 are remarkably small compared to those of the other states. This is because the estimations of x_4 and x_6 are based on direct measurements from the gyroscope with sensor noise, filtered only by the EKF. The results clearly show that the proposed method outperformed the practical KKF-based design approach.

In this paper, we tested proposed models on a relatively simple full car's suspension system. However, as the results from the sensitivity analyses show, the neural network performs well when a sufficient amount of prior knowledge is provided as a form of a training dataset. Likewise, if the prior information contains the key features of nonlinear vehicle dynamics that are important for the neural network training, our proposed neural network is expected to perform well on a more complicated full car suspension system even if there is noise and bias in the sensors. In addition, the proposed method is expected to perform well on other (unseen) road conditions if it is trained with big data, which contains sufficient prior information on such road conditions.

5.3 Online simulations

In Table 6, it can be seen that the performance parameter does not degrade when using the proposed method as it does when using the ground-truth-based controller. Note that the assumption behind this example is that the controller with a better observer performs more

effectively than the controller with a worse observer. Therefore, by differentiating the sensors as described in Fig. 2, we showed that the proposed method is effective when the sensors cannot be sufficiently attached in the practical situations.

5.4 Implementation in a real car

There exist evidences to use trained algorithms from simulations for real situations without any additional training by maintaining a good transferability via domain randomization [21,26]. In a similar manner, the proposed neural network training and utilizing system can be implemented in a real car. On the other hand, by installing the position, velocity, and cheap inertial sensors in a real car, one can collect the data to train the proposed neural network as described in Fig. 2. When training the neural network in a real car, the weights of the trained network in the simulation environment can be utilized as good initial values. This will expedite the training process and may produce a better performance of the proposed method compared to randomly initialized neural networks.

In reality, the condition of the vehicle may change due to manufacturing tolerances or slow mechanical degradation of automotive parts. However, it would be too expensive, if we have to collect big data again whenever the dynamical properties of the vehicle change. To cope with this possible inefficiency, the aforementioned needs for updates in the neural network may be re-formulated into a calibration problem. For example, a conditional neural network [8] can be designed and trained with extensively simulated data such that we may provide a calibration parameter, e.g., θ_c (as an additional input to the neural network) that can be estimated easily using any calibration [22,41] or a parameter estimation technique. Subsequently, the network is trained using the big data along with a variety of θ_c covering the situations that customers may encounter. The fully trained neural network can be then implemented in a production stage as well as in a maintenance by calibrating θ_c .

6 Conclusions

In this paper, we proposed and successfully trained an LSTM-based neural network which reconstructs nearly

perfect states in a full car's nonlinear suspension system, which is not observable, using a big data as a prior distribution. This is the first time that the LSTM-based neural network is designed and implemented for estimating only using cheap inertial sensors such as accelerometers with or without gyroscope sensors in a nonlinear vehicle model. In addition, we showed how to improve the performance through Bayesian filtering by projecting the outputs of the neural network into the vehicle dynamics using the extended Kalman filter. Furthermore, we showed how to utilize the proposed neural network practically by differentiating the selection of the sensors in the training stage and the production stage. To demonstrate the effectiveness of the proposed method, we compared the performance of the state reconstruction of our approach against a conventional kinematic Kalman filter. Our approach showed that the LSTM-based neural network can successfully reconstruct the states, while the conventional kinematic Kalman filter easily diverged. Next, we analyzed the sensitivity of the proposed method with or without gyroscope sensors to identify which sensors are important for the reconstruction of the velocity and position states. Finally, as an illustrative example of controlling semi-active suspension system, we implemented the proposed method to the LQR controller which requires a proper observer to estimate all of the states in a full car dynamics. The results show that the proposed method reconstructs nearly perfect states, and the performance of the implemented LQR controller does not degrade compared to the ground-truth-based controller. As a future plan, we will create more advanced simulation environment by adding nonlinearities and by utilizing more wider variety of road conditions. Subsequently, the overall system will be implemented in a real car and will evaluate whether the proposed method is cost-efficient and accurate when solving the proposed automotive industry problems. Finally, we will also consider the re-formulation of the our problem with changes in mechanical properties near future as discussed in Sect. 5.4.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

References

1. Alamdari, A., Sovizi, J., Krovi, V.N.: Enhanced full-state estimation and dynamic-model-based prediction for road-vehicles. In: ASME 2014 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference. American Society of Mechanical Engineers Digital Collection (2014)
2. Bar-Shalom, Y., Li, X.R., Kirubarajan, T.: Estimation with Applications to Tracking and Navigation: Theory Algorithms and Software. Wiley (2004)
3. Carratù, M., Pietrosanto, A., Sommella, P., Paciello, V.: Velocity prediction from acceleration measurements in motorcycle suspensions. In: 2017 IEEE International Instrumentation and Measurement Technology Conference (I2MTC), pp. 1–6. IEEE (2017)
4. Challa, S., Bar-Shalom, Y.: Nonlinear filter design using Fokker–Planck–Kolmogorov probability density evolutions. *IEEE Trans. Aerosp. Electron. Syst.* **36**(1), 309–315 (2000)
5. Chen, S., Billings, S., Grant, P.: Non-linear system identification using neural networks. *Int. J. Control* **51**(6), 1191–1214 (1990)
6. Chen, Z., et al.: Bayesian filtering: from Kalman filters to particle filters, and beyond. *Statistics* **182**(1), 1–69 (2003)
7. Cheng, Y., Karjala, T.W., Himmelblau, D.M.: Identification of nonlinear dynamic processes with unknown and variable dead time using an internal recurrent neural network. *Ind. Eng. Chem. Res.* **34**(5), 1735–1742 (1995)
8. Codevilla, F., Müller, M., López, A., Koltun, V., Dosovitskiy, A.: End-to-end driving via conditional imitation learning. In: 2018 IEEE International Conference on Robotics and Automation (ICRA), pp. 1–9. IEEE (2018)
9. Davoodabadi, I., Ramezani, A.A., Mahmoodi-k, M., Ahmadizadeh, P.: Identification of tire forces using dual unscented Kalman filter algorithm. *Nonlinear Dyn.* **78**(3), 1907–1919 (2014)
10. Dugard, L., Senname, O., Aubouet, S., Talon, B.: Full vertical car observer design methodology for suspension control applications. *Control Eng. Practice* **20**(9), 832–845 (2012)
11. Gamse, S.: Dynamic modelling of displacements on an embankment dam using the Kalman filter. *J. Spat. Sci.* **63**(1), 3–21 (2018)
12. Gamse, S., Nobakht-Ersi, F., Sharifi, M.A.: Statistical process control of a Kalman filter model. *Sensors* **14**(10), 18053–18074 (2014)
13. García, S.D., Patino, D.: Estimation based on acceleration measures of an active suspension plant. In: 2015 IEEE 2nd Colombian Conference on Automatic Control (CCAC), pp. 1–6. IEEE (2015)

14. Hochreiter, S., Schmidhuber, J.: Long short-term memory. *Neural Comput.* **9**(8), 1735–1780 (1997)
15. Hu, G., Gao, B., Zhong, Y., Gu, C.: Unscented Kalman filter with process noise covariance estimation for vehicular INS/GPS integration system. *Inf. Fusion* **64**, 194–204 (2020)
16. Jeon, S.: State estimation based on kinematic models considering characteristics of sensors. In: *Proceedings of the 2010 American Control Conference*, pp. 640–645. IEEE (2010)
17. Karim, F., Majumdar, S., Darabi, H., Chen, S.: LSTM fully convolutional networks for time series classification. *IEEE Access* **6**, 1662–1669 (2017)
18. Kingma, D.P., Ba, J.: Adam: A method for stochastic optimization. *arXiv preprint [arXiv:1412.6980](https://arxiv.org/abs/1412.6980)* (2014)
19. Martens, W., von Wagner, U., Mehrmann, V.: Calculation of high-dimensional probability density functions of stochastically excited nonlinear mechanical systems. *Nonlinear Dyn.* **67**(3), 2089–2099 (2012)
20. Múčka, P.: Simulated road profiles according to iso 8608 in vibration analysis. *J. Test. Eval.* **46**(1), 405–418 (2017)
21. Muratore, F., Gienger, M., Peters, J.: Assessing transferability from simulation to reality for reinforcement learning. *IEEE Trans. Pattern Anal. Mach. Intell.* (2019)
22. Nagamune, R., Choi, J.: Parameter reduction in estimated model sets for robust control. *J. Dyn. Syst. Meas. Control* **132**(2) (2010)
23. Neal, R.M.: Priors for infinite networks. In: *Bayesian Learning for Neural Networks*, pp. 29–53. Springer (1996)
24. Nirmal, K., Sreejith, A., Mathew, J., Sarpotdar, M., Suresh, A., Prakash, A., Safonova, M., Murthy, J.: Noise modeling and analysis of an imu-based attitude sensor: improvement of performance by filtering and sensor fusion. In: *Advances in Optical and Mechanical Technologies for Telescopes and Instrumentation II*, vol. 9912, p. 99126W. International Society for Optics and Photonics (2016)
25. Park, J., Son, H., Lee, J., Choi, J.: Driving assistant companion with voice interface using long short-term memory networks. *IEEE Trans. Ind. Inf.* **15**(1), 582–590 (2018)
26. Peng, X.B., Andrychowicz, M., Zaremba, W., Abbeel, P.: Sim-to-real transfer of robotic control with dynamics randomization. In: *2018 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 1–8. IEEE (2018)
27. Pesce, V., Silvestrini, S., Lavagna, M.: Radial basis function neural network aided adaptive extended Kalman filter for spacecraft relative navigation. *Aerosp. Sci. Technol.* **96**, 105527 (2020)
28. Poussot-Vassal, C., Sename, O., Dugard, L., Gaspar, P., Szabo, Z., Bokor, J.: A new semi-active suspension control strategy through LPV technique. *Control Eng. Practice* **16**(12), 1519–1534 (2008)
29. Savaresi, S.M., Poussot-Vassal, C., Spelta, C., Sename, O., Dugard, L.: *Semi-active Suspension Control Design for Vehicles*. Elsevier (2010)
30. Smyth, A., Wu, M.: Multi-rate Kalman filtering for the data fusion of displacement and acceleration response measurements in dynamic system monitoring. *Mech. Syst. Signal Process.* **21**(2), 706–723 (2007)
31. Sobczyk, K.: Stochastic nonlinear systems. Existing methods and new results. *ZAMM* **78**(10), 651–661 (1998)
32. Srinivasan, B., Prasad, U.R., Rao, N.J.: Back propagation through adjoints for the identification of nonlinear dynamic systems using recurrent neural models. *IEEE Trans. Neural Netw.* **5**(2), 213–228 (1994)
33. Von Wagner, U.: On non-linear stochastic dynamics of quarter car models. *Int. J. Non-Linear Mech.* **39**(5), 753–765 (2004)
34. von Wagner, U., Wedig, W.V.: On the calculation of stationary solutions of multi-dimensional Fokker–Planck equations by orthogonal functions. *Nonlinear Dyn.* **21**(3), 289–306 (2000)
35. Wang, X., Qin, W., Wang, Y., Cui, N.: Bayesian filtering based on solving the Fokker–Planck equation. *Aircr. Eng. Aerosp. Technol.* **90**(2), 312–319 (2018)
36. Wang, Z., Dong, M., Qin, Y., Du, Y., Zhao, F., Gu, L.: Suspension system state estimation using adaptive Kalman filtering based on road classification. *Veh. Syst. Dyn.* **55**(3), 371–398 (2017)
37. Williams, C.K.: Computing with infinite networks. In: *Advances in Neural Information Processing Systems*, pp. 295–301 (1997)
38. Yi, K., Suk Song, B.: Observer design for semi-active suspension control. *Veh. Syst. Dyn.* **32**(2–3), 129–148 (1999)
39. Zaheer, M., Ahmed, A., Smola, A.J.: Latent lstm allocation joint clustering and non-linear dynamic modeling of sequential data. In: *Proceedings of the 34th International Conference on Machine Learning-Volume 70*, pp. 3967–3976. JMLR.org (2017)
40. Zhang, G., Yu, Z., Wang, J.: Correction of contaminated yaw rate signal and estimation of sensor bias for an electric vehicle under normal driving conditions. *Mech. Syst. Signal Process.* **87**, 64–80 (2017)
41. Zhang, L., Jiang, Z., Choi, J., Lim, C.Y., Maiti, T., Baek, S.: Patient-specific prediction of abdominal aortic aneurysm expansion using Bayesian calibration. *IEEE J. Biomed. Health Inf.* **23**(6), 2537–2550 (2019)
42. Zheng, X., Zaheer, M., Ahmed, A., Wang, Y., Xing, E.P., Smola, A.J.: State space LSTM models with particle mcmc inference. *arXiv preprint [arXiv:1711.11179](https://arxiv.org/abs/1711.11179)* (2017)
43. Zhu, Q., Ishitobi, M.: Chaotic vibration of a nonlinear full-vehicle model. *Int. J. Solids Struct.* **43**(3–4), 747–759 (2006)

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